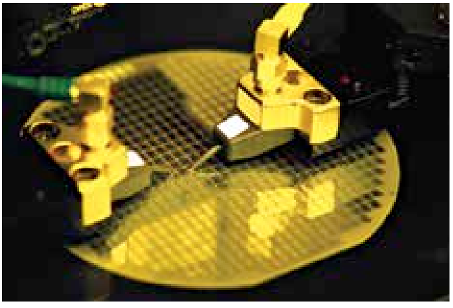
designed with testability features like a built-in self-test or scan chains. Some of the designs which use special analog fab processes feature wafers that are laser trimmed during the testing. This is so that tightly-designed resistance values that the design specifies could be achieved.

A good design would attempt to test and statistically manage corners-which are extreme silicon behavior brought about by high operating temperature and also other extreme processing steps. Most of the designs would be able to cope with a minimum of 64 corners.

Dicing
After testing, a wafer usually gets reduced in thickness. Then, the wafer gets scored and broken into separate dices. This process is called wafer dicing. When it comes to packaging, only the functioning and unmarked chips get packaged, of course.

Packaging
There are three steps involved in packaging-whether ceramic or plastic: mounting the dies, connecting die pads to pins on the package and then the sealing of the dies. To link pads to pins, tiny wires are used. Such wires, traditionally were composed of gold. This brings about a lead frame involving solder-plated copper. Since lead is poisonous, lead-free lead frames are mandatory these days.



Dicing: no game of dice, but a precise science

Yet another packaging technology is chip scale package or CSP. In the case of plastic dual in-line package, it's many times bigger than the die which is inside. However, CSP chips are almost the same size as the die. This means that a CSP could be constructed for every die before the wafer gets diced.

The packaged chips also get retested. This ensures that the chips didn't get damaged while packaging. It also ensures that the die-to-pin interconnect operation was done correctly. Once the retesting is done, a laser would etch the name and numbers of the chip on the package.